

Introduction to Industrial Artificial Intelligence

Homework 03

Objective: Implement the analytical tools and assess the health of the shaft in a rotor-bearing system.

1 Background

See Homework 02 for background information.

2 Problem Statement

Develop both an SOM and SOM-MQE method for assessing the health condition of the shaft based on the provided vibration data.

3 Data Source

The dataset is the same as the dataset for Homework 02. Use the following files from each folder for your assessment.

Training Data:

Healthy – vibration data measured under normal condition (20 observations).

Faulty – (1) vibration data measured under shaft unbalance condition level 1 (1 screw on the disk) (20 observations) **AND** (2) vibration data measured under shaft unbalance condition level 2 (2 screws on the disk) (20 observations).

Testing Data:

Vibration data measured from the testbed with unknown health conditions (30 observations). All health assessment results should come from testing data sets.

4 Suggested Algorithms

Feature Extraction	Extract and add more features from the raw signal (e.g. RMS, peak to peak, standard deviation from the time domain signal, etc.)
Modelling	SOM, SOM-MQE

5 Result Submission

For each **group**, please submit **a report document** and **a 5-10 min presentation (ppt file)** presenting the health assessment result. The submission report and presentation should contain the following sections:

- (1) Introduction
 - a. background information and description of the problem
 - b. raw data description
- (2) Implemented Solution
 - a. brief introduction of the analytics process (sample preparation, feature extraction and selection, application of suggested algorithms)

- b. flow chart of entire solution process
- (3) Results
 - a. figures or tables showing the assessment results for both the SOM and SOM-MQE approaches, including the health label of each test sample and a confusion matrix of the test sample results
- (4) Discussion
 - a. include any observations about the modelling techniques, extracted features, results, etc.
 - b. compare the results with the logistic regression results from Homework 02
- (5) Conclusion
 - a. brief statement concluding the important findings
- (6) Appendix – Codes
 - a. all the programming codes used to achieve the results in the report

6 Appendix - Labels for Testing Data

Use the following to check your results.

Healthy	Unbalance Level 1	Unbalance Level 2
Data Mar-26-14 Time 1533-21	Mar-26-14 Time 1549-12	Mar-26-14 Time 1606-21
Data Mar-26-14 Time 1533-22	Mar-26-14 Time 1549-13	Mar-26-14 Time 1606-22
Data Mar-26-14 Time 1533-23	Mar-26-14 Time 1549-14	Mar-26-14 Time 1606-23
Data Mar-26-14 Time 1533-24	Mar-26-14 Time 1549-15	Mar-26-14 Time 1606-24
Data Mar-26-14 Time 1533-25	Mar-26-14 Time 1549-16	Mar-26-14 Time 1606-25
Data Mar-26-14 Time 1533-26	Mar-26-14 Time 1549-17	Mar-26-14 Time 1606-26
Data Mar-26-14 Time 1533-27	Mar-26-14 Time 1549-18	Mar-26-14 Time 1606-27
Data Mar-26-14 Time 1533-28	Mar-26-14 Time 1549-19	Mar-26-14 Time 1606-28
Data Mar-26-14 Time 1533-29	Mar-26-14 Time 1549-20	Mar-26-14 Time 1606-29
Data Mar-26-14 Time 1533-30	Mar-26-14 Time 1549-21	Mar-26-14 Time 1606-30